

UNIT I – LM 1

INTRODUCTION TO CLOUD COMPUTING

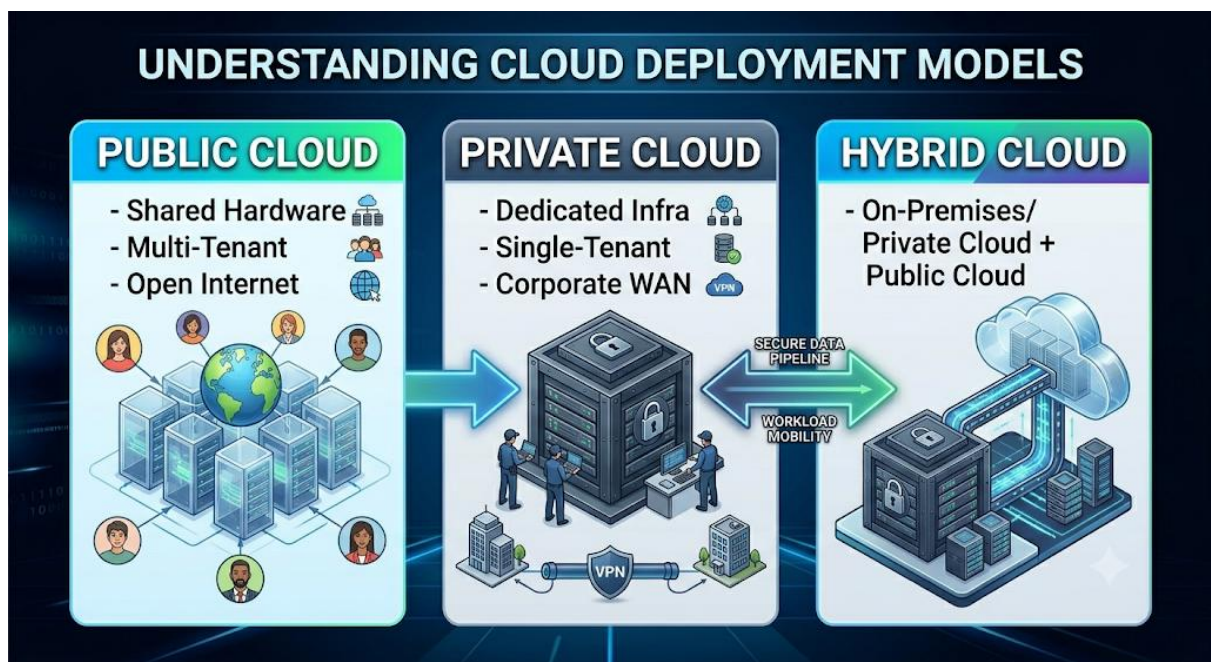
INTRODUCTION TO CLOUD COMPUTING:

Cloud computing has transformed how organizations build, deploy, and scale technology. Instead of owning physical infrastructure, users access computing resources over the internet on demand.

- Eliminates the need for physical servers and upfront hardware costs.
- Enables on-demand access to computing power and storage.
- Supports scalability, flexibility, and cost efficiency.

CLOUD DEPLOYMENT MODELS:

Deployment models specify where the physical computing infrastructure is located and who retains governance rights over that hardware.



PUBLIC CLOUD:

Architecture & Characteristics

The cloud infrastructure is owned and operated by a third-party commercial provider. These providers construct massive data center complexes, partition the physical hardware via hypervisors, and sell computing slices to the general public over the open internet.

Detailed Trade-offs**Pros**

- Minimal CapEx: Zero upfront data center construction expenses.
- High Scalability: Virtually unlimited infrastructure resources available on demand.
- Optimized Resource Allocation: Internal IT engineering resources can focus directly on business software differentiation rather than low-level infrastructure maintenance.

Cons

- Unpredictable Multi-Tenant Latency: Other users on the same physical server can occasionally spike resource usage, sometimes impacting performance ("The Noisy Neighbor Effect").
- Data Sovereignty Realities: Storing sensitive data on shared public infrastructure may run into regulatory hurdles under strict privacy frameworks (like GDPR or HIPAA) if compliance architectures are not carefully designed.
- Long-Term OpEx Cost Curves: For predictable, non-fluctuating workloads operating continuously at scale, the ongoing cumulative cost of public cloud consumption can eventually outpace the cost of owned hardware.

Primary Use Cases

- E-commerce websites subject to significant seasonal traffic spikes.
- Public-facing mobile application backends.
- Big data analytical experiments requiring massive compute power for brief periods.

PRIVATE CLOUD:**Architecture & Characteristics**

The cloud infrastructure is provisioned for exclusive use by a single organization. It can be owned, managed, and operated by the organization itself, a third party, or a combination of both, and may reside physically on-premises or at a co-location data center.

Detailed Trade-offs**Pros**

- Strict Security & Isolation: Completely isolated compute, storage, and networking layers insulate the environment from external multi-tenant exploits.
- Granular Governance: Full architectural freedom to customize hardware configurations, storage controllers, and low-level networking parameters.
- Deterministic Performance: Guaranteed availability of all hardware resources without any performance degradation from outside workloads.

Cons

- **High Upfront Capital Expenditures:** Requires substantial initial capital to build out data center infrastructure, physical machinery, and virtualization licensing.
- **Operational Tail Risk:** The organization retains full responsibility for disaster recovery, physical security, equipment failures, and power outages.
- **Inelastic Scaling Limits:** Once physical capacity is exhausted, provisioning additional resources requires ordering and configuring new hardware, which can introduce significant delays.

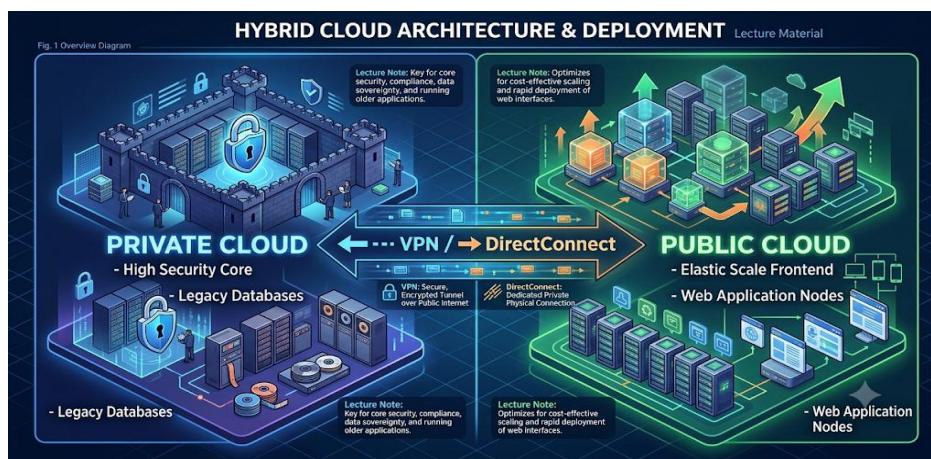
Primary Use Cases

- Large banking institutions and financial clearinghouses managing sensitive transactional records.
- Government agencies and national defense entities requiring isolated networks.
- Healthcare health systems handling highly restricted patient diagnostic systems.

HYBRID CLOUD:

Architecture & Characteristics

A hybrid cloud merges public and private cloud models. The two distinct environments remain separate entities but are bound together by standardized networking technologies and encrypted data pipes. This allows data and application states to move seamlessly between them.



Architectural Mechanisms

- **Hybrid Networking:** Established using secure Virtual Private Network (VPN) tunnels over the internet or via dedicated, private fiber-optic connections (such as AWS Direct Connect or Azure ExpressRoute).
- **Cloud Bursting:** An application deployment configuration where an organization runs its baseline workloads within a private cloud environment. If resource utilization spikes close to capacity limits, the application automatically redirects excess traffic to external public cloud nodes to maintain performance.

Detailed Trade-offs

Pros

- **Optimized Operational Flexibility:** Provides a middle ground between private cloud control and public cloud scalability.
- **Gradual Modernization Pathways:** Organizations can retain legacy core systems in on-premises data centers while deploying modern web components to the public cloud.

Cons

- **Increased Complexity:** Systems engineers must manage disparate software platforms, network routing policies, identity directories, and monitoring systems.
- **Data Integration Challenges:** Synchronizing data states reliably between on-premises databases and remote public cloud endpoints requires careful architectural planning to handle latency and throughput variations.

Primary Use Cases

- Retail businesses keeping their primary user accounts inside an on-premises database while using the public cloud to absorb holiday traffic surges.
- Enterprises running stable baseline operations internally while using public cloud environments for quick software development and testing cycles.

COMMUNITY CLOUD:

Architecture & Characteristics

A community cloud is shared by several organizations that have common requirements, goals, and regulatory concerns (such as security mandates, compliance standards, or shared industry frameworks). It can be managed by the member organizations or a third-party vendor.



Detailed Trade-offs**Pros**

- Shared Financial Burden: Capital costs and operational overhead are distributed across all participating organizations.
- Targeted Compliance Controls: The entire infrastructure can be designed from the ground up to comply with specific industry regulations, avoiding the generic setups of standard public clouds.

Cons

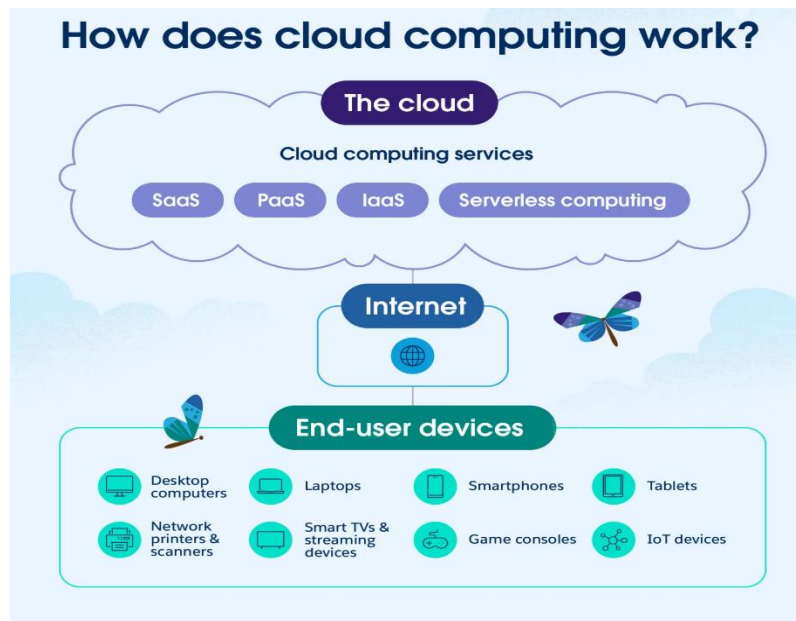
- Collective Governance Constraints: Modifying systemic configurations or shifting priority queues requires multi-party consensus, which can slow down administrative agility.
- Shared Capacity Contention: Computing resources are shared across all member organizations, meaning a traffic spike at one organization can potentially affect the performance of others within the community.

Primary Use Cases

- A consortium of university research departments sharing a supercomputing cluster to run complex scientific simulations.
- Regional credit unions linking their infrastructure together on a shared platform to process payments in compliance with strict financial regulations.

How Does Cloud Computing Work?

- At its core, cloud computing works by shifting the heavy lifting from your local device to a massive, global network of remote data centers. Here is the breakdown of how the tech works behind the scenes:



The Front End: This is what you interact with. It's the user interface—your web browser, a mobile app, or terminal command—used to request a service.

The Back End: This is the invisible powerhouse. It consists of thousands of physical servers, massive data storage arrays, and security systems housed in giant data centers worldwide.

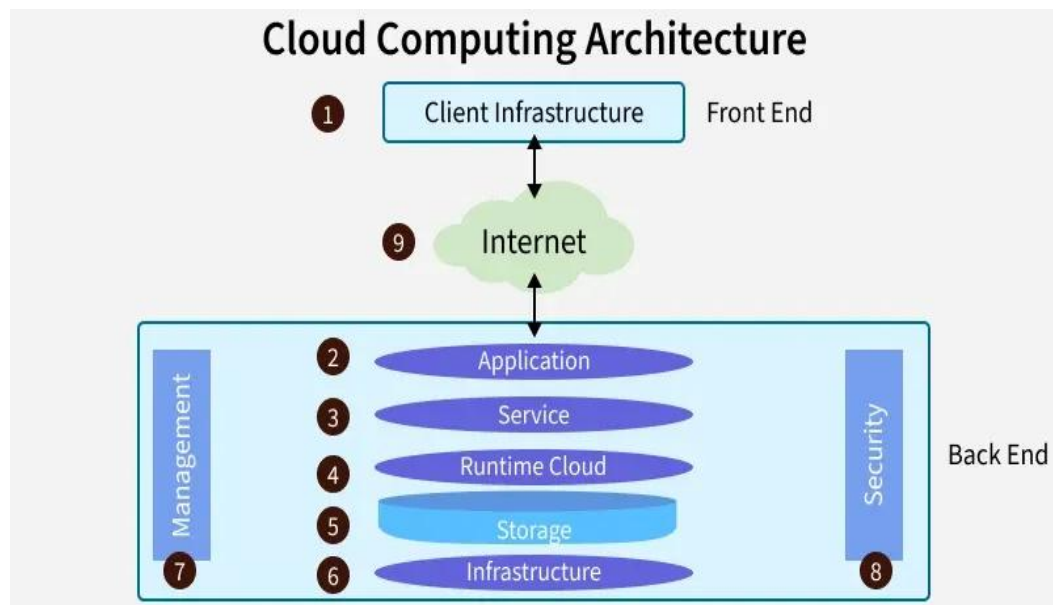
Virtualization (The Secret Sauce): Instead of dedicating one physical computer to one customer, cloud providers use software called a **hypervisor** to slice up one massive physical server into dozens of secure, independent "virtual machines." This is why resource allocation is instant and affordable.

The Internet Highway: High-speed network connections act as the bridge, carrying your request to the data center and sending the processed data back to your screen in milliseconds.

The Electricity Analogy: You don't build a power plant just to turn on a light bulb; you plug into a grid and pay for the electricity you use. The cloud is simply a digital utility grid for computing power.

Cloud Computing Architecture

Cloud computing architecture refers to the components and sub-components required for cloud computing. These components typically refer to:



Cloud Computing Architecture

1. Front End (User Interaction Enhancement)

The User Interface of Cloud Computing consists of 2 sections of clients. The Thin clients are the ones that use web browsers facilitating portable and lightweight accessibilities and others are known as Fat Clients that use many functionalities for offering a strong user experience.

2. Back-end Platforms (Cloud Computing Engine)

The core of cloud computing is made at back-end platforms with several servers for storage and processing computing. Management of Applications logic is managed through servers and effective data handling is provided by storage. The combination of these platforms at the backend offers the processing power, and capacity to manage and store data behind the cloud.

3. Cloud-Based Delivery and Network

On-demand access to the computer and resources is provided over the Internet, Intranet, and Intercloud. The Internet comes with global accessibility, the **Intranet** helps in internal communications of the services within the organization and the Intercloud enables interoperability across various cloud services. This dynamic network connectivity ensures an essential component of cloud computing architecture on guaranteeing easy access and data transfer.

Cloud Service Models

The three primary cloud service models define the level of control and responsibility shared between the cloud provider and the user.

Service Model	Full Form	What Provider Manages	Examples
IaaS	Infrastructure as a Service	Physical servers, storage, networking	AWS EC2, Google Compute Engine, Azure VMs
PaaS	Platform as a Service	IaaS + OS, runtime, middleware, databases	Google App Engine, Heroku, AWS Elastic Beanstalk
SaaS	Software as a Service	Everything including the application	Gmail, Salesforce, Dropbox, Microsoft 365

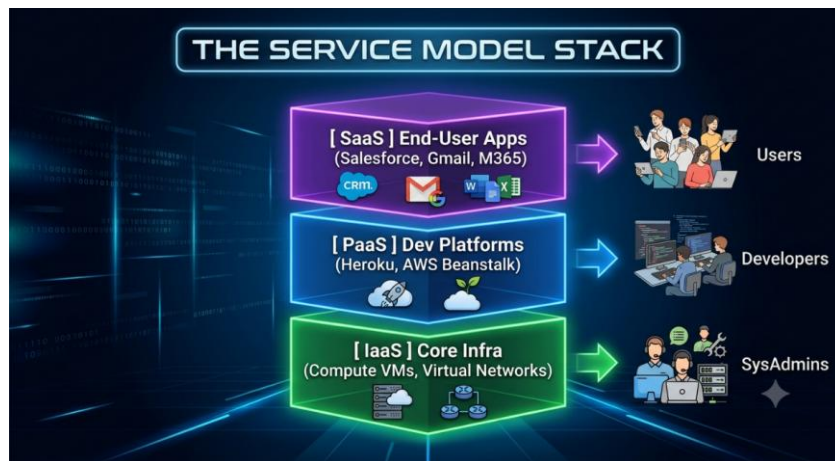


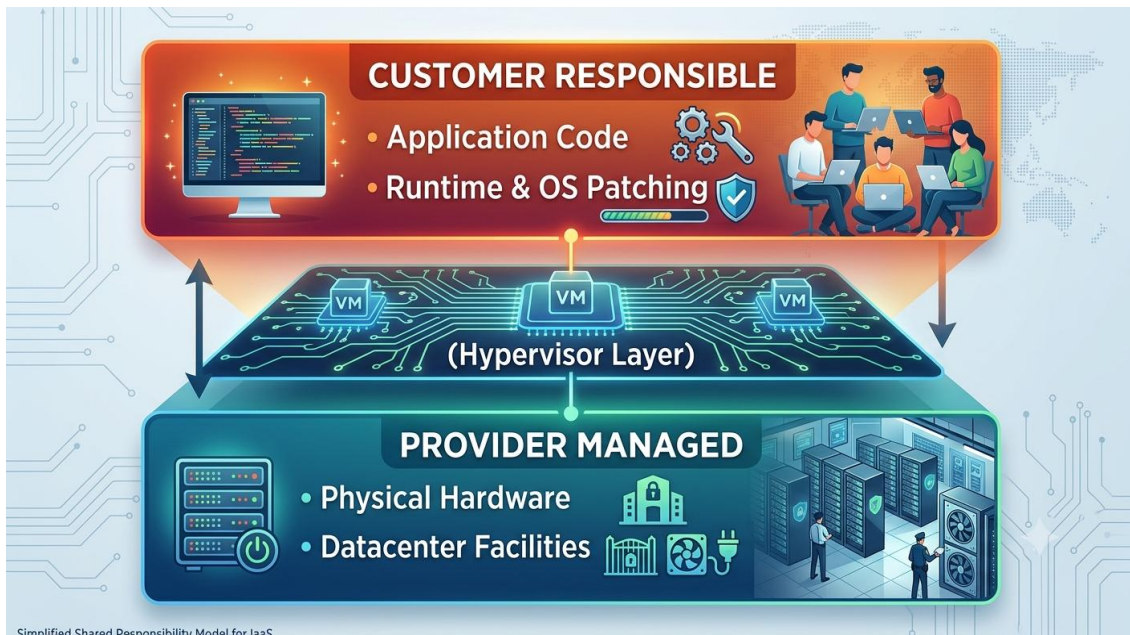
Fig. Cloud Service Models

2.1 IaaS: Infrastructure as a Service

The most fundamental cloud service model. The provider offers virtualized computing infrastructure over the internet.

- User manages: OS, applications, data, middleware, runtime.
- Provider manages: physical servers, storage hardware, networking, and virtualization.
- Best for: Businesses wanting full control over their software stack without hardware costs.
- Examples: Amazon Web Services (EC2, S3), Microsoft Azure VMs, Google Compute Engine.

Analogy (IaaS): IaaS is like renting a plot of land and building bricks. You construct and manage your own house (software), but the land and foundation (infrastructure) is provided.



2.2 PaaS: Platform as a Service

Provides a platform allowing customers to develop, run, and manage applications without dealing with infrastructure complexity.

- User manages: applications and data only.
- Provider manages: OS, runtime, middleware, servers, storage, networking.
- Best for: Developers who want to focus on building apps, not maintaining infrastructure.
- Examples: Google App Engine, Heroku, Microsoft Azure App Services, AWS Elastic Beanstalk.

Analogy (PaaS): PaaS is like renting a fully furnished kitchen. You cook your own meals (write and deploy applications), but all utensils, appliances, and the kitchen itself (platform) are provided.

2.3 SaaS: Software as a Service

The most complete form of cloud service. Software is delivered to end-users directly over the internet, fully managed by the provider.

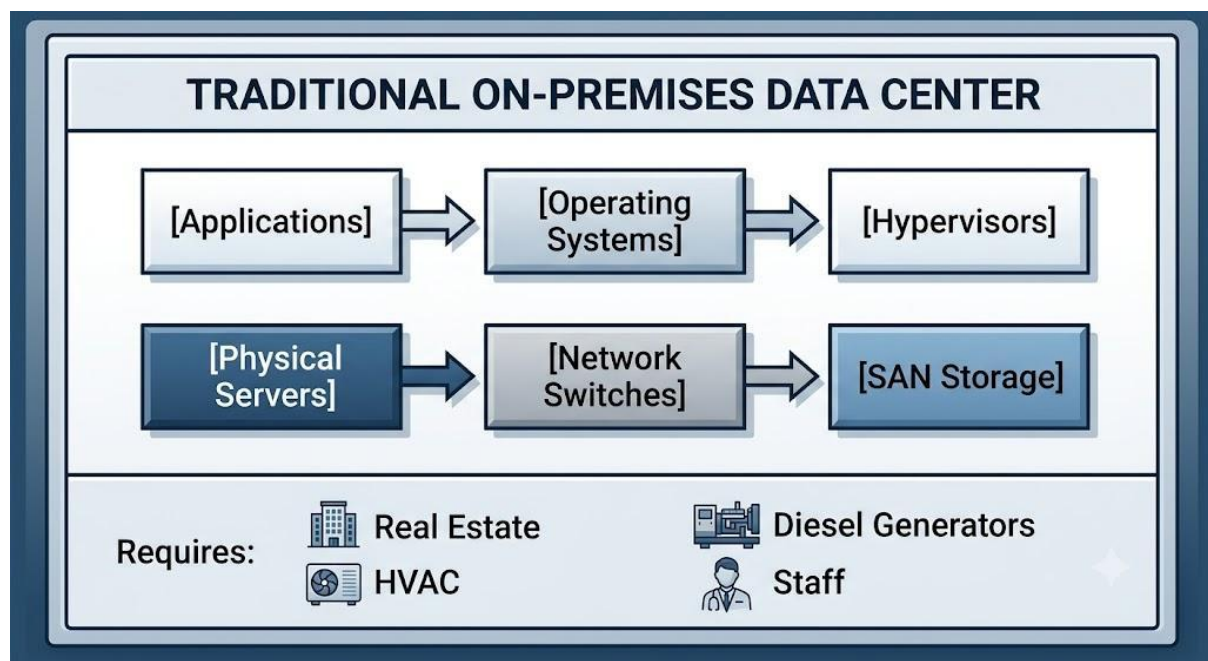
- User manages: only their own data and user-level settings.

- Provider manages: everything including infrastructure, platform, and application.
- Best for: End-users needing ready-to-use applications without installation or maintenance.
- Examples: Gmail, Google Workspace, Salesforce CRM, Dropbox, Zoom, Microsoft 365.

Analogy (SaaS): SaaS is like ordering food from a restaurant. You simply consume the finished product (use the software); cooking, cleaning, and all management is done by the provider.

The Legacy Paradigm: On-Premises Computing

To appreciate the innovation of cloud computing, we must first understand the traditional corporate IT architecture that preceded it. Historically, if an organization required an IT system (such as an enterprise resource planning system, a database, or a web application), it had to construct and maintain its own physical infrastructure.



This traditional approach introduced several operational challenges:

Capital Expenditure (CapEx) Dominance: Procuring hardware required substantial upfront investments. Servers, storage arrays, networking routers, security firewalls, and data center real estate had to be purchased before a single line of code could run.

The Over-Provisioning vs. Under-Provisioning Dilemma: Estimating compute capacity requirements for a new application is notoriously difficult.

Over-provisioning results in idle hardware, leading to wasted budget.

Under-provisioning causes system saturation, server crashes, and poor user experiences during peak traffic.

- **Long Lead Times:** Ordering, shipping, racking, stacking, and configuring physical hardware frequently extended project timelines by weeks or months.
- **Operational Friction:** Internal IT departments functioned as cost centers, spending substantial energy on routine maintenance (such as replacing failed hard drives or tuning backup power systems) rather than focusing on core business logic.